

**SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY COMPUTER SCIENCE
& INFORMATION TECHNOLOGY DEPARTMENT**

FALL 2024

WEB ENGINEERING LAB (CS-311L)

ASSIGNMENT

Semester: 5th
Due Date: 10th January, 2025

Batch: 2022F
Max Marks: 05

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Roll#: 2022F-BCS-152
Section: C

QUESTION:

Create a dynamic website using **HTML, CSS, and JavaScript** with a focus on **DOM manipulation**. Choose one of the following topics:

1. **CGPA Calculator:**

A tool where users can input grades and credit hours to calculate their CGPA.

2. **Standard Calculator:**

A functional calculator that performs:

Basic arithmetic operations: addition, subtraction, multiplication, and division.

Advanced mathematical functions: **sin, cos, tan**, and **square root**.

3. **Interactive Periodic Table:**

A dynamic table displaying chemical elements. Clicking an element reveals detailed information such as atomic number, symbol, and group.

CGPA Calculator

html

```
< !DOCTYPE html >
< html lang = "en" >
< head >
  < meta charset = "UTF-8" >
  < title > CGPA Calculator </ title >
  < link rel = "stylesheet" href = "styles.css" >
</ head >
< body >
  < header >
    < h1 > CGPA Calculator </ h1 >
  </ header >
  < div class= "wrapper" >
    < main >
      < form id = "cgpa-form" >
        < div id = "course-inputs" >
          <!--Courses will be added here -->
        </div>
        <button type="button" id="add-course-btn">Add Course</button>
        <button type="button" id="calculate-btn">Calculate GPA</button>
      </form>
      <div id="result"></div>
    </main>
  </div>
  <script src="script.js"></script>
```

```
</body>
</html>
```

styles.css

```
body {
  margin: 0;
  font-family: Arial, sans-serif;
  background-color: #f5f5f5;
  color: #333;
}

header {
  width: 100%;
  background-color: #4CAF50;
  color: white;
  padding: 15px;
  font-size: 1.5em;
  font-weight: bold;
  text-align: center;
  border-radius: 8px 8px 0 0;
}

.wrapper {
  max-width: 600px;
  width: 90%;
  margin: 20px auto;
  text-align: center;
  background: #fff;
  padding: 20px;
  box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
  border-radius: 8px;
}

main {
  margin-top: 20px;
}

form div {
  margin-bottom: 15px;
}

.course {
  border: 1px solid #ccc;
  border-radius: 6px;
  padding: 15px;
  margin-bottom: 15px;
  width: 100%;
  max-width: 400px;
  margin: auto;
  background: #fafafa;
}

label {
  display: block;
  margin-bottom: 8px;
  font-weight: bold;
  text-align: left;
  color: #555;
}
```

```

input, select {
  width: 100%;
  padding: 10px;
  font-size: 1em;
  border: 1px solid #ccc;
  border-radius: 4px;
  margin-top: 5px;
  box-sizing: border-box;
  outline: none;
}

input:focus, select:focus {
  border-color: #4CAF50;
  box-shadow: 0 0 4px rgba(76, 175, 80, 0.5);
}

#add-course-btn, #calculate-btn {
  display: inline-block;
  padding: 12px 20px;
  font-size: 1em;
  color: white;
  background-color: #4CAF50;
  border: none;
  border-radius: 4px;
  cursor: pointer;
  margin-top: 10px;
  transition: background-color 0.3s ease;
}

#add-course-btn:hover, #calculate-btn:hover {
  background-color: #45a049;
}

#result {
  margin-top: 20px;
  font-size: 1.2em;
  text-align: center;
  border: 1px solid #ccc;
  border-radius: 6px;
  padding: 15px;
  width: 100%;
  max-width: 400px;
  margin: auto;
  background-color: #fafafa;
}

@media (max-width: 600px) {
  header {
    padding: 10px;
    font-size: 1.2em;
  }

  .wrapper {
    width: 95%;
  }

  input, select {
    padding: 8px;
  }

```

```

}

#add-course-btn, #calculate-btn {
  padding: 10px;
  font-size: 0.9em;
}

#result {
  font-size: 1em;
  padding: 10px;
}
}

```

script.js

```

let courseCount = 0;

document.getElementById('add-course-btn').addEventListener('click', function() {
  courseCount++;
  const courseDiv = document.createElement('div');
  courseDiv.className = 'course';
  courseDiv.innerHTML = `
    < h3 > Course ${courseCount}</ h3 >
    < label for= "credit${courseCount}" > Credit Hours:</ label >
    < input type = "number" id = "credit${courseCount}" name = "credit${courseCount}" min = "1"
required >
    < br >
    < label for= "marks${courseCount}" > Marks:</ label >
    < input type = "number" id = "marks${courseCount}" name = "marks${courseCount}" min = "0"
max = "100" >
    < br >
    < label for= "grade${courseCount}" > Or Select Grade:</ label >
    < select id = "grade${courseCount}" name = "grade${courseCount}" >
      < option value = "" > Select Grade </ option >
      < option value = "A" > A </ option >
      < option value = "A-" > A -</ option >
      < option value = "B+" > B +</ option >
      < option value = "B" > B </ option >
      < option value = "B-" > B -</ option >
      < option value = "C+" > C +</ option >
      < option value = "C" > C </ option >
      < option value = "C-" > C -</ option >
      < option value = "D+" > D +</ option >
      < option value = "D" > D </ option >
      < option value = "F" > F </ option >
    </ select >
    < br >
  `;
  document.getElementById('course-inputs').appendChild(courseDiv);
});

function getGradePoint(marks)
{
  if (marks >= 86) return { grade: 'A', point: 4.00 };
  if (marks >= 80) return { grade: 'A-', point: 3.66 };
  if (marks >= 75) return { grade: 'B+', point: 3.33 };
  if (marks >= 70) return { grade: 'B', point: 3.00 };
  if (marks >= 67) return { grade: 'B-', point: 2.66 };
  if (marks >= 63) return { grade: 'C+', point: 2.33 };
  if (marks >= 60) return { grade: 'C', point: 2.00 };
  if (marks >= 57) return { grade: 'C-', point: 1.66 };
  if (marks >= 54) return { grade: 'D+', point: 1.30 };
  if (marks >= 50) return { grade: 'D', point: 1.00 };
  return { grade: 'F', point: 0.00 };
}

function getGradePointFromGrade(grade)

```

```

{
  switch (grade)
  {
    case 'A':
      return 4.00;
    case 'A-':
      return 3.66;
    case 'B+':
      return 3.33;
    case 'B':
      return 3.00;
    case 'B-':
      return 2.66;
    case 'C+':
      return 2.33;
    case 'C':
      return 2.00;
    case 'C-':
      return 1.66;
    case 'D+':
      return 1.30;
    case 'D':
      return 1.00;
    case 'F':
      return 0.00;
    default:
      return null;
  }
}

document.getElementById('calculate-btn').addEventListener('click', function() {
  let totalCredit = 0;
  let totalCP = 0;
  let errorMessage = '';

  for (let i = 1; i <= courseCount; i++)
  {
    const creditInput = document.getElementById(`credit${ i}`);
    const marksInput = document.getElementById(`marks${ i}`);
    const gradeSelect = document.getElementById(`grade${ i}`);

    const credit = parseFloat(creditInput.value);
    const marks = parseFloat(marksInput.value);
    const grade = gradeSelect.value;

    if (isNaN(credit) || credit <= 0)
    {
      errorMessage += `Invalid credit hours for Course ${ i}.< br >`;
    }

    if (!isNaN(marks) && (marks < 0 || marks > 100))
    {
      errorMessage += `Invalid marks for Course ${ i}.< br >`;
    }

    if (!marksInput.value && !grade)
    {
      errorMessage += `Please enter marks or select grade for Course ${ i}.< br >`;
    }

    let point;
    if (marksInput.value)
    {
      if (isNaN(marks))
      {
        errorMessage += `Invalid marks for Course ${ i}.< br >`;
      }
      else

```

```

    {
        const gradeData = getGradePoint(marks);
        point = gradeData.point;
    }
}
else if (grade)
{
    point = getGradePointFromGrade(grade);
    if (point === null)
    {
        errorMessage += `Invalid grade for Course ${ i}.< br >`;
    }
}
else
{
    errorMessage += `Please enter marks or select grade for Course ${ i}.< br >`;
}

if (!isNaN(credit) && credit > 0 && point !== undefined && point !== null)
{
    totalCredit += credit;
    totalCP += point * credit;
}
}

if (errorMessage !== '')
{
    document.getElementById('result').innerHTML = `< span style = "color: red;" >${ errorMessage}</
span >`;
}
else if (totalCredit === 0)
{
    document.getElementById('result').innerHTML = 'No valid courses to calculate GPA.';
}
else
{
    const gpa = totalCP / totalCredit;
    document.getElementById('result').innerHTML = `Your GPA is: ${ gpa.toFixed(2)}`;
}
});

```

GitHub Deployment: <https://faisalbasit.github.io/CGPA-Calculator/>

CGPA Calculator

Course 1

Credit Hours:

1

Marks:

Or Select Grade:

A

Course 2

Credit Hours:

3

Marks:

Or Select Grade:

A

Course 3

Credit Hours:

3

Marks:

Or Select Grade:

A

Course 4

Credit Hours:

3

Marks:

Or Select Grade:

A-

Course 5

Credit Hours:

1

Marks:

Or Select Grade:

A-

Course 6

Credit Hours:

Marks:

Or Select Grade:



Course 7

Credit Hours:

Marks:

Or Select Grade:



Add Course

Calculate GPA

Your GPA is: 3.80